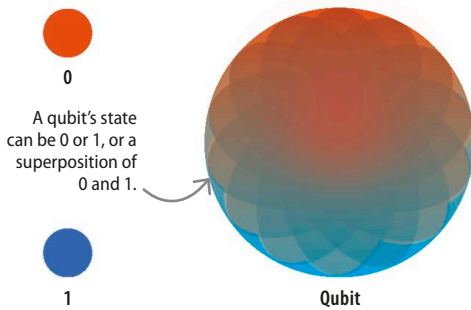


Future programming languages

What sort of languages would be needed to program entirely new types of computer? Quantum computers use the principles of quantum physics to do calculations, which would take an impossibly long time using normal computers. They're currently in the early stages of development, but Quantum Computing Language (QCL), based on C, has already been created for them.



◁ Qubit

A bit in a quantum computer is known as a quantum bit, or qubit. It can be in one of three possible states: 0, 1, or a state where it is both 0 and 1 at the same time. The last state is known as a quantum superposition.

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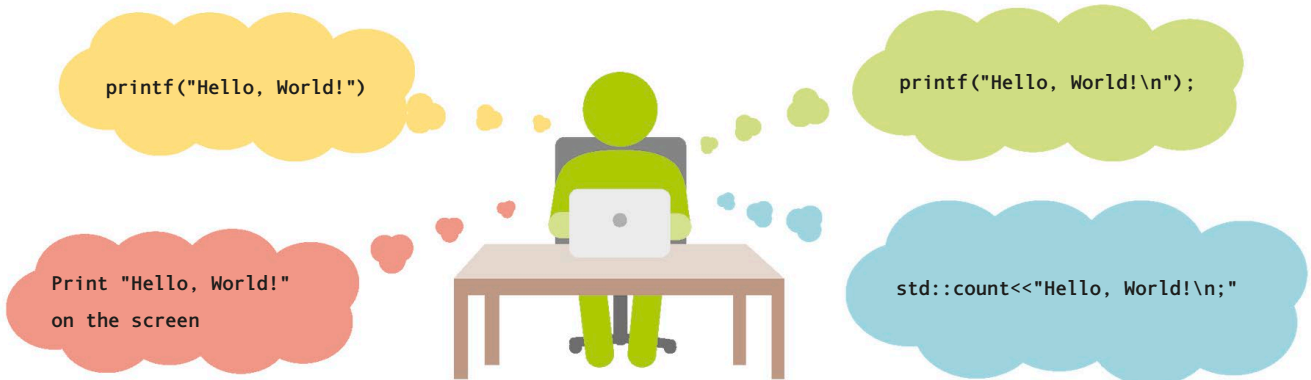
Molecular computing

Biological engineers at Massachusetts Institute of Technology (MIT), USA, recently developed a programming language that enabled them to construct biochemical circuits made from DNA. These circuits are placed in biological cells, enabling the cells to react to their environment in specific ways.



A universal language?

It's unlikely that programming languages will move towards a situation where there is one language used for everything. Just like with physical tools, each language is specifically designed to have particular strengths. Machine learning – the ability for computers to learn new things without being specifically programmed – is likely to have an effect on programming in the future, as programmers will make use of tools that have this ability.



△ Transferable skills

Programming involves taking an algorithm and expressing it using instructions a computer can understand. It's a skill that is transferable from one programming language to another. The traditional "Hello, World!" greeting can be written in different programming languages, but have the same result.

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Machine learning languages

While machine learning may reduce the need for traditional programming skills, the machine learning systems themselves will have to be written by programmers. The most popular languages used to create learning systems include Python, the programming language R, and Java.

